THE INERTIA FIBRED CATEGORY IS EQUIVALENT TO THE DIAGONAL SELF PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7036-7093, following the statement at 6996-7034.

Editorial note. Author writes 'Some details omitted' at 7092 covering well-definedness of the quasi-inverse on morphisms and naturality of the displayed isomorphism; both are routine checks of data displayed in-span.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 7092 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- 2-fibre product of fibred categories over C. Lemma lemma-2-product-fibred-categories-over-C of this chapter, invoked at proof lines 7037-7042. The explicit construction of S x_{S'} S and of the diagonal self product whose objects are triples (x, y, (iota, kappa)).
- Fibred equivalence invariance. Lemma lemma-fibred-equivalent of this chapter, invoked at proof lines 7038-7039. Deduces (2) (fibredness of the inertia category) from (1) (the equivalence).

Source reference. Original lemma 02XQ: sources/stacks-categories/pinned-original-source.tex, source lines 6775–6775. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 042G: sources/stacks-categories/pinned-original-source.tex, source lines 6696–6696. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $p : \mathcal{S} \to \mathcal{C}$ and $p' : \mathcal{S}' \to \mathcal{C}$ be fibred categories. Let $F : \mathcal{S} \to \mathcal{S}'$ be a 1-morphism of fibred categories over $\mathcal{C}$. Consider the category $\mathcal{I}_{\mathcal{S}/\mathcal{S}'}$ over $\mathcal{C}$ whose*

(1) *objects are pairs (x, α) where $x \in \mathrm{Ob}(\mathcal{S})$ and $\alpha : x \to x$ is an automorphism with $F(\alpha) = id$,*

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(2) *morphisms* $(x, \alpha) \to (y, \beta)$ *are given by morphisms* $\phi : x \to y$ *such that*

$$\begin{array}{ccc} x & \xrightarrow{\phi} & y \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\beta} \\ x & \xrightarrow{\phi} & y \end{array}$$

commutes, and

(3) *the functor* $\mathcal{I}_{\mathcal{S}/\mathcal{S}'} \to \mathcal{C}$ *is given by* $(x, \alpha) \mapsto p(x)$.

Then

(1) *there is an equivalence*

$$\mathcal{I}_{\mathcal{S}/\mathcal{S}'} \longrightarrow \mathcal{S} \times_{\Delta,(\mathcal{S}\times_{\mathcal{S}'}\mathcal{S}),\Delta} \mathcal{S}$$

in the $(2,1)$*-category of categories over* $\mathcal{C}$*, and*

(2) $\mathcal{I}_{\mathcal{S}/\mathcal{S}'}$ *is a fibred category over* $\mathcal{C}$.

Proof. Note that (2) follows from (1) by Lemmas 02XQ and 042G. Thus it suffices to prove (1). We will use without further mention the construction of the 2-fibre product from Lemma 02XQ. In particular an object of $\mathcal{S} \times_{\Delta,(\mathcal{S}\times_{\mathcal{S}'}\mathcal{S}),\Delta} \mathcal{S}$ is a triple $(x, y, (\iota, \kappa))$ where x and y are objects of $\mathcal{S}$, and $(\iota, \kappa) : (x, x, \mathrm{id}_{F(x)}) \to (y, y, \mathrm{id}_{F(y)})$ is an isomorphism in $\mathcal{S} \times_{\mathcal{S}'} \mathcal{S}$. This just means that $\iota, \kappa : x \to y$ are isomorphisms and that $F(\iota) = F(\kappa)$. Consider the functor

$$I_{\mathcal{S}/\mathcal{S}'} \longrightarrow \mathcal{S} \times_{\Delta,(\mathcal{S}\times_{\mathcal{S}'}\mathcal{S}),\Delta} \mathcal{S}$$

which to an object (x, α) of the left hand side assigns the object $(x, x, (\alpha, \mathrm{id}_x))$ of the right hand side and to a morphism ϕ of the left hand side assigns the morphism (ϕ, ϕ) of the right hand side. We claim that a quasi-inverse to that morphism is given by the functor

$$\mathcal{S} \times_{\Delta,(\mathcal{S}\times_{\mathcal{S}'}\mathcal{S}),\Delta} \mathcal{S} \longrightarrow I_{\mathcal{S}/\mathcal{S}'}$$

which to an object $(x, y, (\iota, \kappa))$ of the left hand side assigns the object $(x, \kappa^{-1} \circ \iota)$ of the right hand side and to a morphism $(\phi, \phi') : (x, y, (\iota, \kappa)) \to (z, w, (\lambda, \mu))$ of the left hand side assigns the morphism ϕ. Indeed, the endo-functor of $I_{\mathcal{S}/\mathcal{S}'}$ induced by composing the two functors above is the identity on the nose, and the endo-functor induced on $\mathcal{S} \times_{\Delta,(\mathcal{S}\times_{\mathcal{S}'}\mathcal{S}),\Delta} \mathcal{S}$ is isomorphic to the identity via the natural isomorphism

$$(\mathrm{id}_x, \kappa) : (x, x, (\kappa^{-1} \circ \iota, \mathrm{id}_x)) \longrightarrow (x, y, (\iota, \kappa)).$$

Some details omitted. □